

FREE

Your Cells Are Starving for Creatine

If you're not eating this type of diet you need a supplement.



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JUN 23, 2024



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Creatine is like your second mitochondria. Or, the mitochondria's chief of staff. Or its co-pilot.

Your mitochondria make ATP so you can see clearly, hear accurately, digest your food, power your brain, show off your shiny skin, lift heavy things, and perform your best at the challenges you face. They do that all with the help of creatine.

Creatine is responsible for spreading the impact of mitochondrial ATP production into the general area of the cell known as the cytosol, and into every organelle outside the mitochondria.

While it is *more* important in cells with high ATP requirements, variable ATP requirements, and long distances between mitochondria and the source of ATP utilization, it is still incredibly important in every cell.

There is no point in optimizing your mitochondria if you don't also optimize your creatine.

Many people may believe that the high muscle creatine stores that athletes achieve with creatine supplements are “unnatural” and something not achievable until creatine supplements were available.

Here, I argue that nothing could be further from the truth. *Every muscle fiber* wants to be *exactly* as rich in creatine as achieved with creatine supplementation.

All of your cells want to be rich in creatine. Your brain is *dying* to be this rich in creatine. Your muscles are *starving* to be this rich in creatine.

It is completely natural to be this rich in creatine, yet most of us in the modern era who don't supplement just *aren't that optimized*.

The creatine we require to be optimized is likely etched deep into our beings by our ancestral consumption of one to two pounds of meat per day. When red and rare, one pound can give the dose that saturates tissue stores. When white and well done, two pounds may be required.

But can we synthesize enough creatine ourselves when all the precursors in place?

Here we examine that question.

But first, a brief review of creatine's lesser known benefits.

This is educational in nature and not medical or dietetic advice. See [terms](#) for additional and more complete disclaimers.

Far More Than a Performance Enhancer

Creatine is by far the most well-studied, well-supported nutritional supplement for athletic performance.

It's about a lot more than that, though. 90-95% is found in skeletal muscle, but here is what the rest of it is doing:

- In sperm, creatine provides the power used to flip the tail and swim up the vaginal canal. **This suggests that creatine is important to male fertility.**
- In our eyes, creatine provides the power needed to translate light and darkness into the electrical signals that generate vision once integrated within our brains. **This suggests that creatine is important to healthy vision.**
- In the hairs of our inner ear, creatine provides the power needed for high-sensitivity hearing, balance, and equilibrium. In mice, creatine supplementation protects them from noise-induced hearing loss. Thus, **creatine is important to healthy hearing.**
- In our stomach, creatine provides the power to pump the stomach acid needed to digest our food. In the intestines, creatine provides the power needed to absorb nutrients. Our intestines have large finger-like projections known as villi that provide the enormous surface area needed for nutrient absorption, and the cells at the tips of the villi need to be replaced every few days. Creatine provides the large amount of energy needed to replace these cells on such a quick schedule. All of this suggests that **creatine is very important to digestion and gastrointestinal health.**
- In our skin, creatine provides the power for the production of the keratin that lines the outer surface. It powers hair growth, too. It even powers the production of sebum, which keeps our skin water-proof and lubricated. When scientists study isolated skin cells, topical application of creatine protects them from the damage induced by ultraviolet light. Creatine also provides the power for wound healing. All of this suggests that **creatine can help maintain vibrant, healthy, youthful skin.**
- **Our brains** are only two percent of our bodyweight, but consume twenty percent of our energy. This huge energy demand is largely provided by creatine.

I have covered and referenced this material before:

- [Consuming Creatine in Foods and Supplements](#)
- [Creatine: Far More Than a Performance Enhancer](#)

Three to five grams per day over four weeks or longer is sufficient to saturate muscle stores and is well studied and widely used in the context of athletic performance.

This dose has also been shown to improve symptoms of [major depression](#) in women.

Recently, a [single dose of 20 grams](#) was shown to improve cognitive function during sleep deprivation.

[Other research](#) has shown 20 grams of creatine for seven days to improve cognitive function in older adults or in young individuals subjected to hypoxia. 20 grams of creatine per day for six months in traumatic brain injury patients reduced amnesia and improved a large number of metrics of recovery.

Few Adverse Effects of Creatine

Creatine is the most well studied supplement in existence, and the most popularly used among athletes, and has nearly nothing documented in the literature as predictable adverse effects.

However, anecdotally, people have reported elevated creatinine, insomnia, cramps, bloating, constipation, water retention, hair loss, irritation, anger, anxiety, lightheadedness during lifting, aggravation of restless leg syndrome, irritation of asthma, bloody noses, headaches, heart palpitations, and changes in heart rate.

These are likely a result of imbalances with electrolytes and amino acids that have simple solutions, and are covered in [Handling Creatine Side Effects](#).

Creatine Supplements Bring Tissue Creatine to the Natural, Optimal Level

Many people seem to believe that the increase in muscular creatine stores achieved with supplementation is unnatural.

This is contradicted by the earliest studies on the matter.

In [an early 1974 study](#), Harris and colleagues measured muscle creatine stores of 81 athletically untrained subjects between the ages of 18-30, sex not mentioned, and found an average of 124.4 millimoles per kilogram of dry matter. The range was not clearly described, but the standard deviation was 11.2. In a typical bell curve distribution, this implies 95% of people have muscular creatine content between 102 and 146.8.

In a [groundbreaking 1992 study](#), the same group studied increases in muscular creatine after different dosing regimens of creatine supplementation at rest and with exercise.

The most salient point from this study was that when the individual changes in each subject were plotted against baseline muscle creatine content, the effect of every regimen was simply to bring everyone, as the authors described in the abstract “closer to the upper limit of the normal range.”

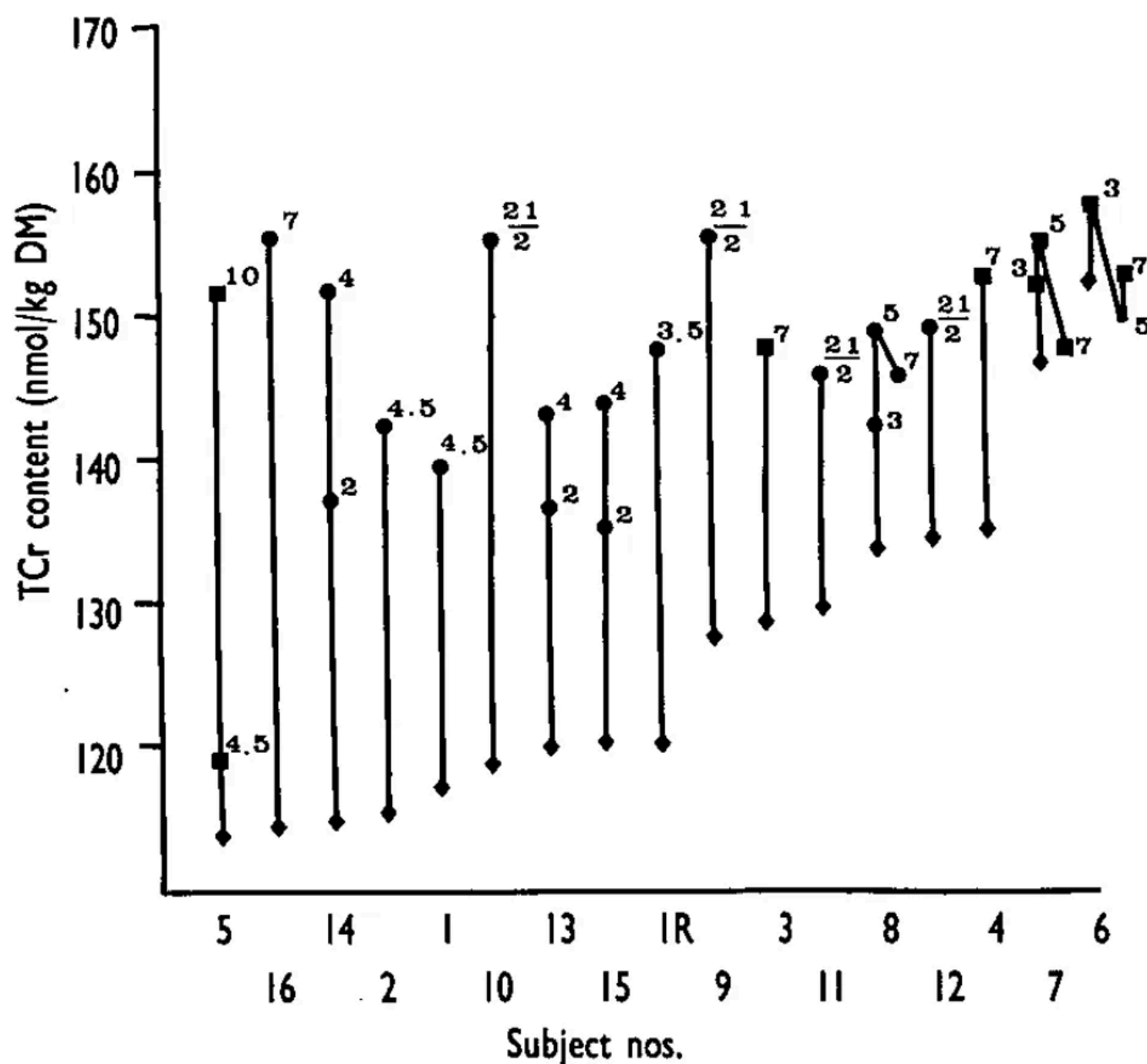


Fig. 2. TCr content of the quadriceps femoris before (◆) and after (■, ●) supplementation with Cr.H₂O. Supplementation rates were as follows: 4 × 5 g for 4.5 days (subject nos. 1 and 2), 7 days (subject nos. 3 and 4) and 10 days (subject nos. 5); 6 × 5 g for 7 days (subject nos. 6–8) with biopsies on days 3, 5 and 7, and on alternate days for 21 days (subject nos. 9–12). Also included are the results from the control leg of five subjects who performed 1 h of strenuous exercise per day with the collateral leg. Supplementation rates were in this case: 4 × 5 g for 3.5 days (subject no. 1R); 6 × 5 g for 4 days (subject nos. 13–15) with biopsies on days 2 and 4, and 7 days (subject no. 16). Subjects have been arranged in order of increasing initial TCr content. Numbers on the Figure denote the days of supplementation at the time of the biopsy, ■, Female subjects; ●, male subjects.

The vertical bars are big, indicating big increases, when the person had baseline creatine content below 120. If their baseline creatine content was in the 140s, they hardly moved at all. Everyone converged on the 140-160 range.

All of Harris's dosing regimens were permutations of 20 grams a day in multiple doses for different durations, or in some cases alternating days.

In a subsequent classic 1996 study by another group, "[Creatine Loading in Men](#)," Hultman and colleagues showed that creatine stores top out in the 140-160 range

whether one does a 20-gram loading phase or takes the slow and steady approach of 3 grams per day.

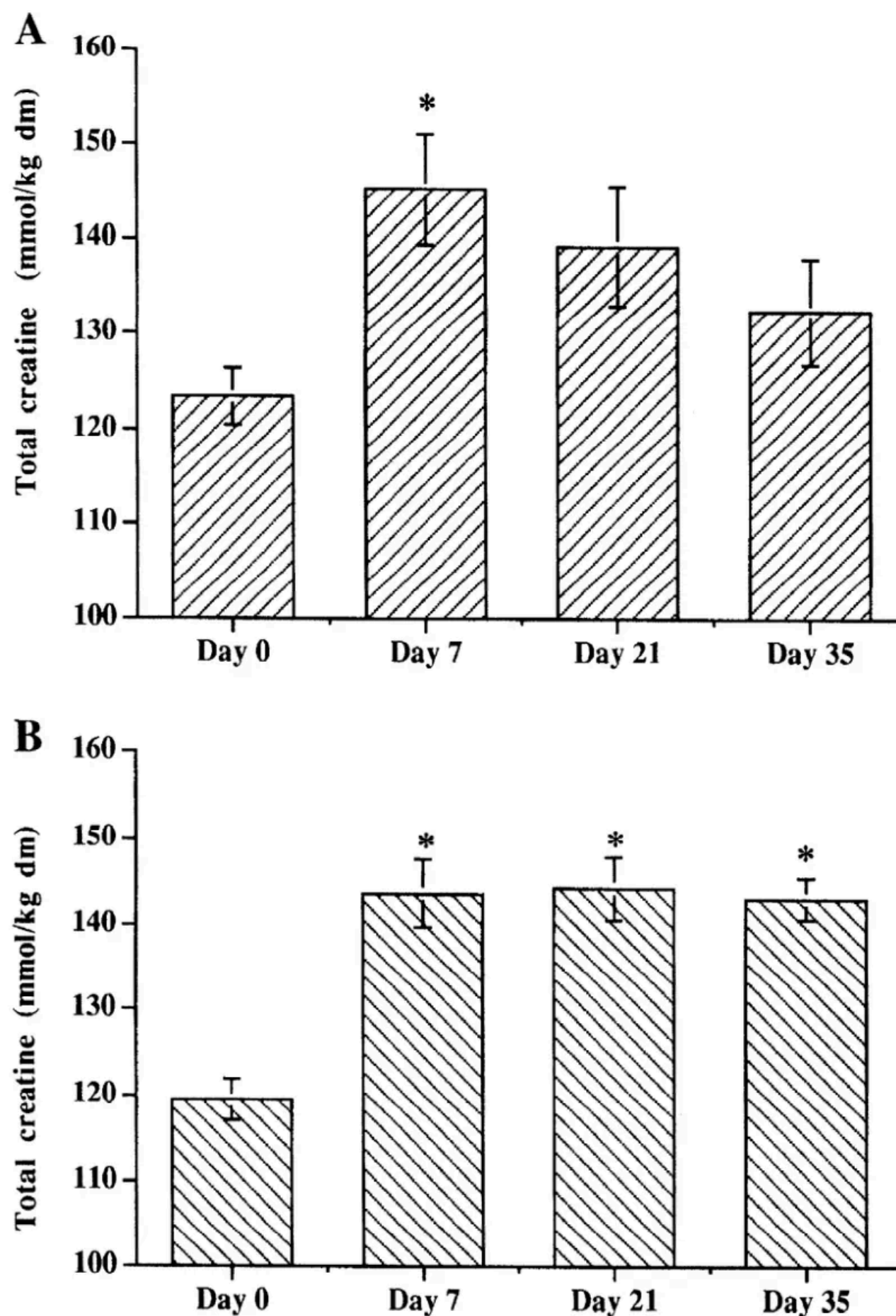


Fig. 1. A: muscle total creatine concentration in experimental group 1 ($n = 6$). All subjects ingested 20 g of creatine for 6 consecutive days, and muscle biopsy samples were obtained before ingestion (*day 0*) and on *days 7, 21, and 35*. B: muscle total creatine concentration in experimental group 2 ($n = 9$). All subjects ingested 20 g of creatine for 6 consecutive days and thereafter ingested creatine at a rate of 2 g/day for the next 28 days. Muscle biopsy samples were taken before ingestion (*day 0*) and on *days 7, 21, and 35*. Values are means $\pm$ SE. dm, Dry mass. * Significantly different from *day 0*, $P < 0.05$.

The error bars are standard errors, not standard deviations, so they visually underestimate the range, and the ranges were not reported. However, focusing on the means, the most salient point is that every regimen peaks at a little over 140.

The top graph shows 20 grams of creatine for six days followed by withdrawal of creatine. The bottom graph shows the same loading dose followed by a two gram per day maintenance dose.

A third group took 3 grams a day for the entire period and experienced the same rise in creatine stores, shown in the last column of this table:

Table 2. *ATP, PCr, free Cr, and total Cr concentrations recorded over 28 days in experimental group 3*

	ATP	PCr	Free Cr	Total Cr
Day 0	23.63 ± 0.70	76.89 ± 2.64	44.87 ± 3.64	121.76 ± 3.35
Day 14	23.51 ± 0.76	83.46 ± 2.05‡	53.08 ± 2.33†	136.54 ± 3.14†
Day 29	23.13 ± 0.53	86.00 ± 1.69‡	56.05 ± 2.55†	142.04 ± 3.18†

Values are means ± SE for 9 subjects in *group 3*. *Group 3* consumed Cr at a rate of 3 g/day for the initial 28 days. Significantly different from *day 0*: †*P* < 0.01; ‡*P* < 0.001.

Note that, throughout the studies, there is a slow path and a rapid path to get to a muscular creatine content in the 140-160 range, and that's it.

The capping out is thought to be from the downregulation of creatine transporters in muscle after they reach the saturation point.

Yet that makes it clear that they *want* to reach that point. They *could* have downregulated the transporters when they had reached average creatine status, but they didn't.

It seems evident from this that muscles *want* to be in the 140-160 range, and that some people get close without dietary creatine and others require a supplement.

Muscle doesn't try to be in the 100s or the 120s. It doesn't try to be in the 200s. As long as the supply is sufficient, muscle converges on a creatine content in the 140-160 range. The variation is larger at baseline, and it shrinks as muscle converges on the creatine stores it wants.

The [average person](#) synthesizes 0.97 grams of creatine per day and eats 1-2 grams per day. Supplementing 21 grams per day drops the rate of synthesis by 30% to 0.68 grams per day. The fact that 2-3 grams of supplemental creatine per day over the long-term can saturate muscle stores suggests that the total requirement between food, synthesis, and supplements to saturate muscle stores is 4-6 grams.

The actual creatine demand will be a function of lean body mass, but does not vary by sex after accounting for women having less lean body mass than men.

While brain creatine does not always change to the same proportion as muscle creatine upon supplementation, they [change in the same direction](#), so the conditions that lead to muscle saturation of creatine stores are likely leading to the optimization of whole-body creatine status and provide an index of how much we need for whole-body creatine optimization.

So, if we need this much why don't we synthesize this much?

How We Synthesize Creatine

Creatine is synthesized in two steps. First, arginine is bound to glycine to produce guanidinoacetate. Second, guanidinoacetate is methylated to produce creatine.

The first reaction is catalyzed by arginine:glycine amidinotransferase (AGAT).

The second is catalyzed by guanidinoacetate methyltransferase (GAMT).

Reviews commonly describe creatine synthesis as an interorgan process where the kidneys produce the bulk of the guanidinoacetate, the liver converts guanidinoacetate to creatine, and 90-95% of the creatine is taken up by muscles.

However, this is likely an oversimplification. For example, a [2021 paper](#) pointed out that this “traditional dogma of creatine metabolism” was contradicted by several papers produced over 40 years suggesting skeletal muscle can synthesize a considerable portion of its own creatine.

[AGAT is expressed](#) in not only the kidney, but also the pancreas, liver, heart, brain, and testes.

A [2007 paper](#) found that in rats the kidney synthesizes almost all the guanidinoacetate, but in humans the kidney only makes 11-12% of it.

Nevertheless, the liver is the most important site of methylation so is likely the disproportionate producer of the body's creatine.

The AGAT enzyme is [thought to be](#) “regulated tightly,” with thyroid hormone and growth hormone increasing it, and creatine inhibiting it in a negative feedback loop.

If the negative feedback from creatine is strong, then one could argue that the average creatine status in the population is the way it is because we make just the amount of creatine we need.

That would indicate that saturating muscle stores with creatine supplements is more than we need, yet it would make it seem bizarre that muscles keep their creatine transporters up at full volume until they reach that higher point.

The reality is that these enzymes are not *tightly* regulated. We see this in the [above-cited paper](#) where 21 grams of creatine per day decreases the rate of synthesis by only 30%. That's the loss of a third of a gram of synthesized creatine for every 21 grams being sucked up into tissue from supplements. That's *loose* regulation.

Granted, it might only be so loose because most of us are so far away from optimal creatine status. But it is clear that the reason our creatine status is lower than we would get from supplementation is not because our body is trying to keep it that way.

So, then, what is stopping us from making more creatine?

Is Methylation Limiting for Creatine Synthesis?

45% of methylation is used for creatine synthesis, 45% is used for phosphatidylcholine synthesis, and the remaining 10% is used on dozens of processes, some prominent ones being the regulation of gene expression, the intracellular degradation of histamine, and the regulation of dopamine and other neurotransmitters.

It is rather clear from this alone that, in the average person, methyl supply has to limit creatine synthesis. If we 2.2x'd our creatine synthesis while holding methyl supply constant, we would not have any methyl groups left for anything else.

And we would still only reach 2 grams of creatine synthesized when we need a total of 4-6 grams.

However, methylation being limiting is contradicted by a human trial where 2 grams of betaine (trimethylglycine or TMG) taken for ten days had [no effect](#) on creatine stores.

The rate of methylation [is estimated](#) to be 16.7–23.4 millimoles per day in young people weighing 70 kilograms, and 15.5–21.7 millimoles per day in older adults weighing 70 kilograms. If we take the upper end, this will provide a conservative estimate of whether the TMG should have been expected to increase creatine synthesis if methyl groups were limiting.

2 grams of betaine is 17 millimoles. Each molecule has three methyl groups, so this is 51 millimoles of methyl groups, 2.2 times the upper estimate of the average daily methyl flux, and 4.8 times the flux devoted to creatine synthesis.

The extra methyl groups were plenty.

Thus, it *has* to be the case that average methyl supply imposes a strict limitation on creatine synthesis, yet it is clear from the betaine trial that *something else* is *even more limiting*.

What Is Standing In the Way of Creatine Synthesis?

In my 2017 podcast, [Living With MTHFR](#), I followed Reed's 2015 paper, [Mathematical Analysis of the Regulation of Competing Methyltransferases](#) to argue that creatine synthesis is very sensitive to methyl supply, while other functions such as DNA methylation are not.

This was supported by modeling the different enzymes having different binding affinities for methyl groups, such that processes we want to be independent of methyl supply like DNA methylation bind very tightly, and processes that we want to sacrifice when methyl supply is low like creatine synthesis bind more loosely. This modeling suggested that in the fasting state most methylation reactions run normally, but

creatine synthesis declines. In the fed state, the influx of methionine allows creatine synthesis to catch up, and any extra methyl groups are mopped up by glycine.

The rationale for this is simple. We don't want genes to necessarily be regulated by the fasting-feeding cycle when they need to be regulated by many other things, we don't want to be sneezing between meals from undermethylation of histamine, and we don't want to become pathological ruminators in the fasting state from undermethylation of dopamine.

By contrast, creatine synthesis can take the hit for a few hours. The average person has 120 grams of creatine in their body, with 1-2% of it degrading to creatinine and leaving in the urine each day. If we assume that we have enough creatine, all that has to happen to stay alive and reasonably healthy is to have the credits and debits in the creatine account balance out at the end of the month. We just need to make sure that over the long-haul we still have on average 120 grams of creatine in our body on any given day.

While this makes a lot of sense, there are multiple studies that contradict this prioritization:

- [In 1971](#), Keshavarz and Fuller showed in male broiler chicks in their first two weeks of life that extra arginine increased muscle creatine stores, but extra glycine and methionine did not. However, extra arginine decreased growth and increased loss of creatine in the urine. These negative effects were alleviated by supplying extra methyl donors in several ways: methionine, choline, betaine, or the combination of B12 and folic acid. This suggested that the supply of arginine was the main driver of creatine synthesis, but that if methyl group supply was insufficient to handle the arginine load, methyl depletion would occur, causing much of the creatine to be wasted by going into the urine and, more importantly, hurting the growth of the chicks. **In this study, young chicks sacrificed their growth to increase their creatine stores.**
- [In 1986](#), experiments with human breast cancer cells showed that when methionine supply was low, the cells sacrificed their ability to methylate histones in the regulation of gene expression just so they could keep synthesizing creatine. When they were deprived of arginine and glycine, histone methylation was restored. **While this could represent an abnormality of breast cancer cells, synthesizing creatine took priority over regulating gene expression.**
- [In 2015](#) it was shown in 14-16-week-old Yucutan miniature pigs that guanidinoacetate supplementation increased muscle creatine stores but decreased the synthesis of choline, the methylation of proteins, and the liver's concentration of S-adenosyl-methionine. **This indicates that creatine synthesis took priority over everything else.**
- However, [in 2016](#) a study found that in miniature pigs that were 4-9 days old, phosphatidylcholine synthesis took priority over creatine synthesis. **This suggests phosphatidylcholine synthesis may take priority over creatine synthesis in certain contexts. My guess is that this only applies when cholinergic nervous system development is prominent, which is very limited in most animals but in humans extends into the fourth year of life.**

The first three studies above all suggest that arginine supply is the main driver of creatine synthesis. However, if cells under certain conditions prioritize the conservation of methyl groups, then extra methyl groups would be required to match creatine synthesis to the arginine supply.

This seems to be the case in newborn miniature pigs, and again my suspicion is that this is because the demand for phosphatidylcholine is high in the neonatal period because cholinergic neurons may still be developing.

[A 2021 study](#) in miniature pigs 9-11 days old showed that doubling the supply of arginine and methionine above normal requirements increased kidney creatine 2.5-fold and liver creatine more than 5-fold, while seeming to slightly increase brain creatine and have no impact on muscle creatine. Supplementing creatine or guanidinoacetate plus methionine had a similar impact, while guanidinoacetate without methionine had no effect. **In this case, arginine or guanidinoacetate drove creatine synthesis only when extra methyl groups were available.**

A [2007 study in rats](#) found that in rats fed the standard AIN-93G diet, thought to be optimized for the nutritional requirements of the rat, infusion of glycine did nothing to increase guanidinoacetate synthesis, while infusion of arginine increased it 150% and infusion of citrulline, which generates arginine in the kidney, nearly doubled it.

Notably, the above studies show that regulation of the enzymes *follows* the increase of creatine status to the saturated state we associate with supplementation.

For example:

- [In humans](#), 21 grams of creatine per day for five days, which is known to saturate creatine stores, causes a 30% decrease in creatine synthesis, which is not remotely sufficient to stop creatine stores from saturating.
- In [miniature pigs](#), doubling arginine and methionine over “adequate” intakes doubled guanidinoacetate production while cutting AGAT activity in half. AGAT activity went down *because* the supply of the substrates was sufficient to double its product despite the lower activity of the enzyme.
- [In rats](#), 4 grams per kilogram bodyweight creatine per day for two weeks decreased AGAT activity by 86%, but this is like a human taking 280 grams of creatine a day adjusting for bodyweight alone or taking 1,680 grams a day after adjusting for surface area. So this was orders of magnitude more than needed for saturation and after two weeks AGAT was still operating with residual activity.

We Have Our Answer!

Clearly what limits creatine synthesis most is the supply of arginine.

And if we supply enough arginine, the methyl groups *must* become limiting because the normal flux of methyl groups is not sufficient for optimal creatine synthesis even if it were used for nothing else.

Under certain conditions, glycine could probably become limiting, but glycine was never limiting in the studies reviewed above.

If 5 grams is the amount of creatine we need to synthesize to hit our optimal concentration with synthesis alone, due to their different weights, this would require 6.6 grams of arginine.

The [average adult in the US](#) consumes about five grams of arginine per day. About 1.3 grams is used for creatine synthesis, leaving 3.7 to be used for other purposes. An additional 5.3 grams of arginine would be needed to meet the need for creatine through synthesis alone, bringing the average intake up to 10.3 grams.

Methyl supply would then become limiting. Two grams of TMG supplies 4.8 times the methyl groups devoted to synthesizing one gram of creatine per day, so adding this on top of usual methyl donor intake would bring the total methyl supply sufficient to synthesize the 5 total grams of creatine.

Calculating the supply of methyl groups needed from the folate/B12 pathway becomes very complicated, because the methyl group comes primarily from glucose, which is used to synthesize serine, which then donates a carbon to folate during the synthesis of glycine. Alternatively, in the presence of TMG, one of the three methyl groups passes through the betaine:homocysteine methyltransferase (BHMT) enzyme, while the other two are funneled through folate and B12 as an alternative to the use of glucose.

Supplementing with methylfolate to get the methyl group that comes preformed in the supplement is a joke. Methylfolate is much bigger than TMG, and therefore weighs more, and it only has one methyl group instead of three. It would take 20 grams to get sufficient methyl supply for creatine synthesis, and that is not safe.

Nevertheless, you need sufficient folate and B12 to use the methyl groups of TMG. See my [Custom Nutrient Targets](#) for those two vitamins.

To get the methyl groups from methionine would require 5.2 grams of methionine above and beyond the 2.1 grams of methionine typically consumed in the US.

Some amount of TMG could be replaced by choline as well, but the base choline requirement is already 450-900 milligrams per day.

My [Choline Database](#) gives food sources of choline and TMG. Getting 2 grams of the sum of choline and betaine over the basic requirement would require eating a lot of quinoa, eggs, liver, and/or wheat germ.

It is possible but very difficult to hit these targets on a vegan diet. You could get 2.5 grams of choline plus TMG from 500 grams of quinoa, and that would give you 5.5 grams of arginine. Add 500 grams of tofu, and you hit 10.2 grams of arginine. You're already at 2358 Calories, so you probably don't have that much room for very high nutrient-density, low-calorie vegetables to round out micronutrient intakes, and you'd need some supplements, especially B12.

On a vegetarian diet with eggs and dairy, you could take advantage of higher methionine levels, but you'd need to eat even more total protein to get enough

arginine.

Meat and fish are by far and away the easiest ways to support creatine status with whole foods. As covered in my database, [Consuming Creatine in Foods and Supplements](#), the rule of thumb is that “red and rare” requires one pound per day, while “white and well done” requires two pounds per day.

That gets you the preformed creatine that supports optimal status *and* it provides higher levels of arginine and methionine. Eggs, liver, legumes, dark greens, and collagenous tissues on the side provide plenty of choline, folate, and glycine to assist in the synthesis of additional creatine. Since the creatine is consumed pre-formed in the meat, the other nutrients are simply available in excess to provide an extra window of safety.

While there are vegan ways to hack increasing creatine synthesis, these have not been shown in human trials to work, and they require planning your entire diet around trying to support creatine synthesis. It is far easier to just take a supplement.

The Bottom Line

Our cells are starving for creatine and are not satiated until they reach the creatine status reached by most people only through supplementation.

They do *not* keep themselves at “normal” creatine status through regulation. Creatine synthesis is limited first by the supply of arginine, and once arginine is supplied it is limited by methyl group supply. In some people with low glycine status, it may also be limited by glycine.

There is nothing “natural” about having average unsupplemented creatine status. Your body hates it.

Achieving optimal creatine status without supplements requires one pound of “rare and red” meat or fish per day, which can include salmon, or two pounds of “white and well done” meat or fish per day.

Vegans, people who require low-arginine diets to manage herpes outbreaks, people who have impairments in the urea cycle or specific pathways of amino acid metabolism, people who feel better on lower-meat diets, or people who for whatever reason do not want to eat 1-2 pounds of meat per day should supplement with 3-5 grams per day of Creapure creatine monohydrate.

Handling Creatine Side Effects

Despite the necessity for everyone of maximizing their creatine stores, some people report elevated creatinine, insomnia, cramps, bloating, constipation, water retention, hair loss, irritation, anger, anxiety, lightheadedness during lifting, aggravation of restless leg syndrome, irritation of asthma, bloody noses, headaches, heart palpitations, and changes in heart rate.

These are addressed in this article:

[Click Here to Read “Handling Creatine Side Effects”](#)

What’s Your Experience With Creatine?

Let me know in the comments!

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ROBERT M HANSEN MD Jun 23, 2024 ...

♥ Liked by Chris Masterjohn, PhD

Be aware that serum creatinine (a measure of renal function, used to calculate estimates glomerular filtration rate aka eGFR) will increase about 10% on 5 grams per day of creatine. Your physician will be alarmed and concerned that kidney function has deteriorated. But this is an "artifact" of supplementation, the kidneys are not deteriorating. Most physicians are clueless, so stop creatine supplementation about 1 week before lab tests.

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11 replies by Chris Masterjohn, PhD and others



Jennifer Depew, R.D. Jun 24, 2024 ...

♥ Liked by Chris Masterjohn, PhD

Creatine has been helping my mother and I since I read that it would support methylation. I have been using about 5 grams in a serving of bean soup for her (Alzheimer's) and she has been more functional. Thanks Chris.

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